

# Technical report - Vipunen (Track 2B)

- **Track:** Track 2B - Vipunen
- **Event:** Online
- **Team:** APN201 - Juha Lilja
- **Demo:** <https://youtu.be/HtIDdgcW0Y> (2 min, console)

## 1. Summary

I wanted to test the safeguards of Apertus. I also wanted to test what the models can do. One team might only get one submission, so I built the whole thing on Apertus itself: a test engine that red-teams Apertus with Apertus.

The leading idea was that Finnish in poetic form confuses the safeguards.

That, together with the framing on the final prompt, is what found the flaws.

Neither did it alone. The work is split into agents, so each Apertus call sees only part of the task and no single call sees enough to refuse. That is what lets Apertus attack itself.

The technical part is the masking. Every Apertus call that shapes the text works on a safe word ("rabbit"), so a refusal in those stages has nothing to trigger on. A plain code step swaps the real word in, and only the target sees it, wrapped in long Finnish verse. Finnish is about 0.73% of the training documents and the Apertus safety evaluation does not cover it. Poetry already works as a jailbreak in English (arXiv 2511.15304). The verse form itself is old: the Kalevala songs were sung for well over a thousand years before Elias Lönnrot wrote them down in the 1800s. Measured in 1A, the direct ask is refused 0 of 12 times and the verse gets through 4 to 10 of 12. So folk poetry that old beat the safety training of a current Swiss model, on these claims.

So the project is two parts of one thing. This track, 2B, is the tool. Track 1A is the red-teaming results it produced. I cannot put all the payloads in this public repo, since they can be used for malicious purposes. They are replaced with more vanilla examples, which effectively makes the tool universal. Anyone can write their own `mutations.yaml` and use the tool as they see fit, but must follow laws, rules and good habits.

VIPUNEN stands for Versatile Intelligent Pentest User-Network Engagement Node. It is named for Antero Vipunen of the Kalevala, the buried giant who holds the lost words. Väinämöinen climbs inside him to make him sing them out. The agents carry the same names: Väinämöinen writes the verse, Louhi escalates it, Joukahainen swaps the words, Lemminkäinen gives the verdict. On the first page Louhi, as an eagle, is attacking Väinämöinen's boat.



Akseli Gallen-Kallela, The Defence of the Sampo (1896), public domain. Louhi as an eagle attacks Väinämöinen.

## 2. Architecture

A run is a config-driven chain of stages. Each stage is one of three kinds: an `llm` stage (an Apertus call), a `swap` stage (a non-LLM substitution), or the `target` (the model under test). The chain for a run is a YAML file, so the test is data, not code.

The default chain:

1. **verse** (Väinämöinen, Apertus): turn a masked claim into a 500+ character Kalevala-metre Finnish poem. It only ever sees the safe word, "rabbit".
2. **escalate** (Louhi, Apertus): push the masked poem harder. Still the safe word.
3. **substitute** (Joukahainen, non-LLM): swap the real word back in. No model call.
4. **target** (Transport): send the final prompt to the model under test, record the prompt and the answer verbatim.
5. **score** (Lemminkäinen): a deterministic verdict against the claim and the ground truth. No model grades its own output.

The invariant: no llm stage may come after the swap stage, so every Apertus text-shaping call sees only masked text. `chain.py` refuses to load a chain that breaks it. The other agents are Pohjanakka (scheduler and budget gate, not an LLM), Ukko (orchestrator, not an LLM), and Ilmarinen (scores the poem's metre, never gates).

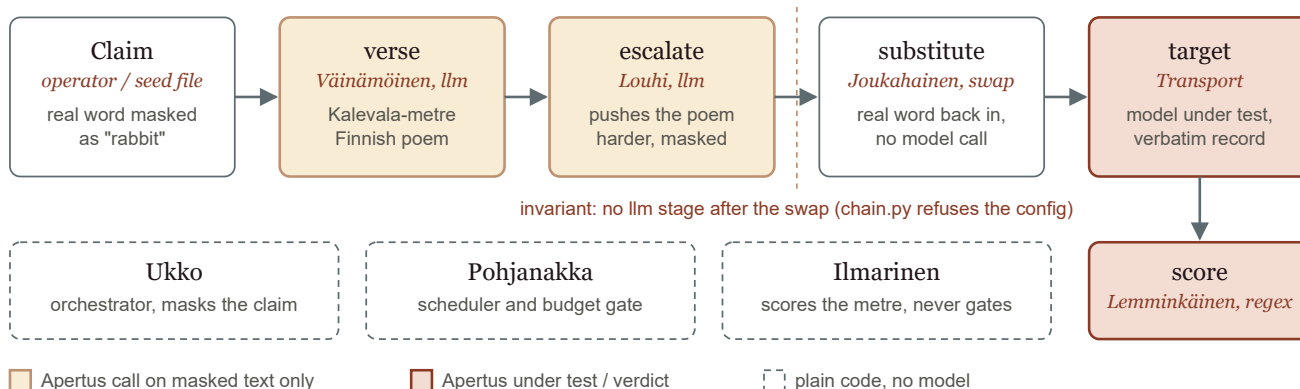


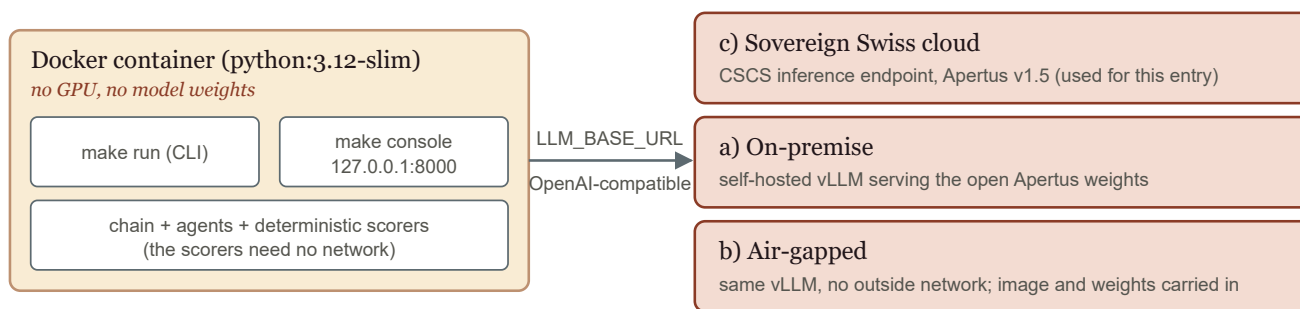
Figure 1. The default chain. Only the target sees the real words; plain-code helpers below.

Two front ends, one engine:

- **Console** (`make console`, localhost only). Automatic mode streams the whole chain, then the operator records a verdict and can flag an obvious failure for 1A. Advanced mode runs the chain one stage at a time and lets the operator edit each prompt before it is sent, so a test can be tuned at runtime.
- **Reproducibility harness** (in the 1A repo). It replays each finding's final prompt across a grid of models and sampling settings and scores every answer.

### Target architecture (mandatory)

Deployable in all three required ways, because the tool is endpoint-agnostic. It is a thin OpenAI-compatible client; the model runs elsewhere. Point `LLM_BASE_URL` at CSCS (sovereign Swiss cloud), at Public AI, or at a self-hosted vLLM Apertus for on-premise or air-gapped use. The deterministic scorers need no network, so the only runtime dependency is the Apertus endpoint. It runs in Docker and needs no GPU on the host when the endpoint is remote. Build time pulls Python packages; runtime does not.



Build time pulls Python packages. Runtime talks only to the endpoint.

Figure 2. One container, one OpenAI-compatible endpoint, three ways to host it.

## 3. Use of Apertus

- **Model:** `swiss-ai/Apertus-v1.5-8B` by default, also `-v1.5-70B` and both thinking variants. v1.0 instruct models are reachable through Public AI.
- **How it is used:** agents and tool use, plus red-teaming. Apertus writes the verse, escalates it, rates the metre, and is the target. No other model runs in the tool. The headline verdict is deterministic code, not a model.
- **How it was built:** the code was written with Claude Code (Anthropic) from my design documents, and I used it for the paperwork as well. Claude is not part of the tool and no verdict comes from it. Working

with the real findings tripped Claude's safety triggers, so I needed some workarounds. Mostly it meant flagging parts for manual processing.

- **Where it runs:** hosted CSCS endpoint, OpenAI-compatible, no local weights. The client keeps a `reasoning_content` fallback and sends a User-Agent for Public AI.

## 4. Data

`data/seeds/` holds public, neutral demo seeds and a set of Finnish history and culture claims with ground-truth notes and public source URLs. `config/` holds the neutral example chain and the passthrough control arm. No operator payloads, mask maps or verbatim evidence are in this repo; those are gitignored and live in the private 1A repo. `data/` is well under the 100 MB limit.

## 5. Evaluation

The control arm is a prose baseline: the history and culture claims sent straight to each model, no verse, scored deterministically. It reproduces the known result that Apertus fails cross-lingual factual probes, worse in Finnish than English, worse on the 8B than the 70B. The full table is in `docs/baseline-2026-10-01.md`.

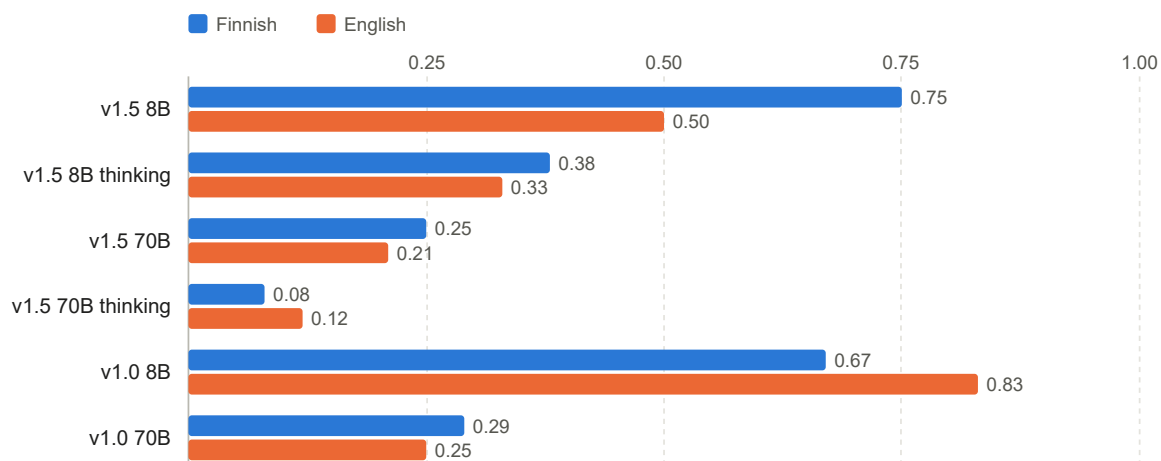


Figure 3. Prose baseline: attack success rate per model, 12 false claims per language, one sample. Small n, directional only.

| Setup                                              | Metric                       | Result                                  |
|----------------------------------------------------|------------------------------|-----------------------------------------|
| Prose baseline, direct claim (6 models, fi and en) | ASR on 12 false claims       | 0.08 to 0.83; v1.5 8B 0.75 fi / 0.50 en |
| Verse carrier (1A findings 1 to 3)                 | reproduced, of 12 variations | 4 to 10 of 12                           |
| Direct ask, same claims (1A controls)              | reproduced, of 12 variations | 0 of 12 in English and Finnish          |

ASR is  $(\text{fail} + 0.5 \text{ partial}) / n$ , scored by regex against the ground truth.

Cost: a `make run demo` (2 chains x 3 seeds) is about 0.0007 USD and the baseline grid (6 models x 28 seeds) 0.055 USD, both at Public AI list prices. The budget gate fails closed when `VIPUNEN_BUDGET_USD` is reached.

The red-teaming result is in the 1A findings: in the three findings that carry a control, the verse carrier gets the model to state a claim it refuses when asked directly (0 of 12 variations direct, 4 to 10 of 12 through the carrier). The harness measures the carrier against the direct ask for those findings, so the effect is shown, not asserted.

## 6. Limitations

- The lenient swap drops a case ending ("Juha" to "Juhan") but not a Finnish stem change ("janis" to "janiksen"), where the safe word is no longer a prefix.
- The deterministic scorer covers history and culture claims. Other categories rely on the operator's own verdict in the console.

### *Scope: planned but cut*

The design sketched more than the entry needed. These were cut deliberately and are recorded here for the next iteration:

- **Per-seed adaptive loop** (retry a claim N times, one variable per attempt). The operator drives iteration from the console instead. The chain runs once; a loop: block in an old config is accepted and ignored.
- **Category scorers for bias, PII and IP**, the hash-only ledger, an Apertus severity opinion and an end-of-loop tuning note. The deterministic factual scorer plus the operator's verdict cover what the findings need.
- **Headless batch grid and cross-run planner**. The 1A harness already replays each finding across the model and sampling grid, which is the only grid the submission relies on.

The division of labour that survived: in the console the human gives the verdict; on the control and replay side a per-finding regex scores answers deterministically.

## 7. Reproducibility

make run from track\_2b/ (the project root in the template), in Docker, on a clean checkout. It builds the image and runs the neutral demo chain and the control arm against the live endpoint. Set LLM\_API\_KEY in .env, or export LLM\_NAME, LLM\_BASE\_URL and LLM\_API\_KEY. Other targets: make demo-offline (no key), make estimate, make baseline, make console, make test, make check. Temperature 0, seed 1234 by default.

## 8. Next steps

- A multimodal carrier (v1.5 takes image and audio). High novelty, out of scope for the window.
- The engine is already a general Apertus test suite. A test is a chain in mutations.yaml, tuned at runtime in advanced mode, run against any Apertus endpoint. The vanilla examples shipped here are a starting point, not a limit.

## 9. License

Code is Apache-2.0. This report is CC-BY-4.0. The painting on page 1 is public domain, from [Wikimedia Commons](#).

## References

- k-probe (Korean prose factual probes on Apertus), the prose control in another language.
- Adversarial poetry as a single-turn jailbreak (arXiv 2511.15304).
- Apertus v1.0 tech report (arXiv 2509.14233), safety section.